

Instructions. Write your answers to the following questions and submit them on Canvas before midnight on the due date. Your answers must be organized by question number and the question clearly marked. Include your first and last name, the course (CPSC-321), and the assignment number at the top of your write up. Your answers must be submitted as a *single* PDF file called **hw-1.pdf**. This assignment is worth 15 points.

For Part A and C below, your answers must:

- Include each relation's name, attributes, primary key(s), foreign keys, and any additional candidate keys (as appropriate).
- Use the standard relational schema notation $R(a_1, a_2, a_3, \dots)$.
- State each relation's primary key by underlining the primary key attributes. List any additional candidate keys separately.
- For each foreign key, include, at a minimum, the source table attribute(s), the target table, and the target table attributes.

Part A: Relational Schema Design

For each question below, design one or more relations to store the described information. Additionally, for each relation you define, provide *at least three* example rows. The questions build on each other to form an airline database.

1. [2pts] US airports have a unique three-letter identifier assigned by the FAA (e.g., GEG, SEA, LAX), the airport name (e.g., "Spokane International", "Seattle-Tacoma International"), and the city and state the airport is located in.
2. [2pts] Airlines have a unique two-character airline code (e.g., AS, WN, UA), an airline name (e.g., "Alaska Airlines", "Southwest Airlines"), the FAA identifier of the airline's primary hub airport (see question 1), and the year the airline was founded.
3. [3pts] Flight routes consist of the airline that operates the route, a flight number, the departure airport, the arrival airport, and the distance of the route in miles. A flight route is identified by its airline code and flight number (no two routes have the same combination).
4. [3pts] Passenger and booking information. Each passenger has a unique passenger id, a first name, a last name, and a unique email address. A booking records that a passenger is booked on a flight route for a specific travel date, along with the seat assignment and the price paid. A

passenger can fly the same route on different dates, but can only be booked once per route per date.

Part B: Relational Schema Diagram

5. [3pts] Draw a single relational schema diagram that contains all of your relation definitions from questions 1 through 4. Be sure to show primary keys, foreign keys, and the relationships between tables. Refer to the lecture notes for the conventions used in relational schema diagrams (these are *not* ER diagrams). You can use one of the tools suggested in the lecture notes to draw the diagram.

Part C: Redundancy

6. [2pts] Consider the following relational schema for storing aircraft maintenance records.

```
MaintenanceRecord(aircraft_id, maintenance_date, mechanic_id, aircraft_model, manufacturer,  
                  mechanic_name, work_performed, parts_cost, mechanic_cert_level)
```

A “good” relational design typically minimizes *redundant information*, i.e., the same information repeated across multiple rows. For example, in the table above, every aircraft has one model and one manufacturer. So the same **aircraft_model** and **manufacturer** values are repeated in every maintenance record for that aircraft. This is redundant because the model and manufacturer are really facts about the aircraft, not about a specific maintenance record.

- (a). What other redundant information is in the table? That is, what other values are unnecessarily repeated across rows?
- (b). How would you redesign this table to remove the redundancy? (Hint: break it into smaller, linked tables.)